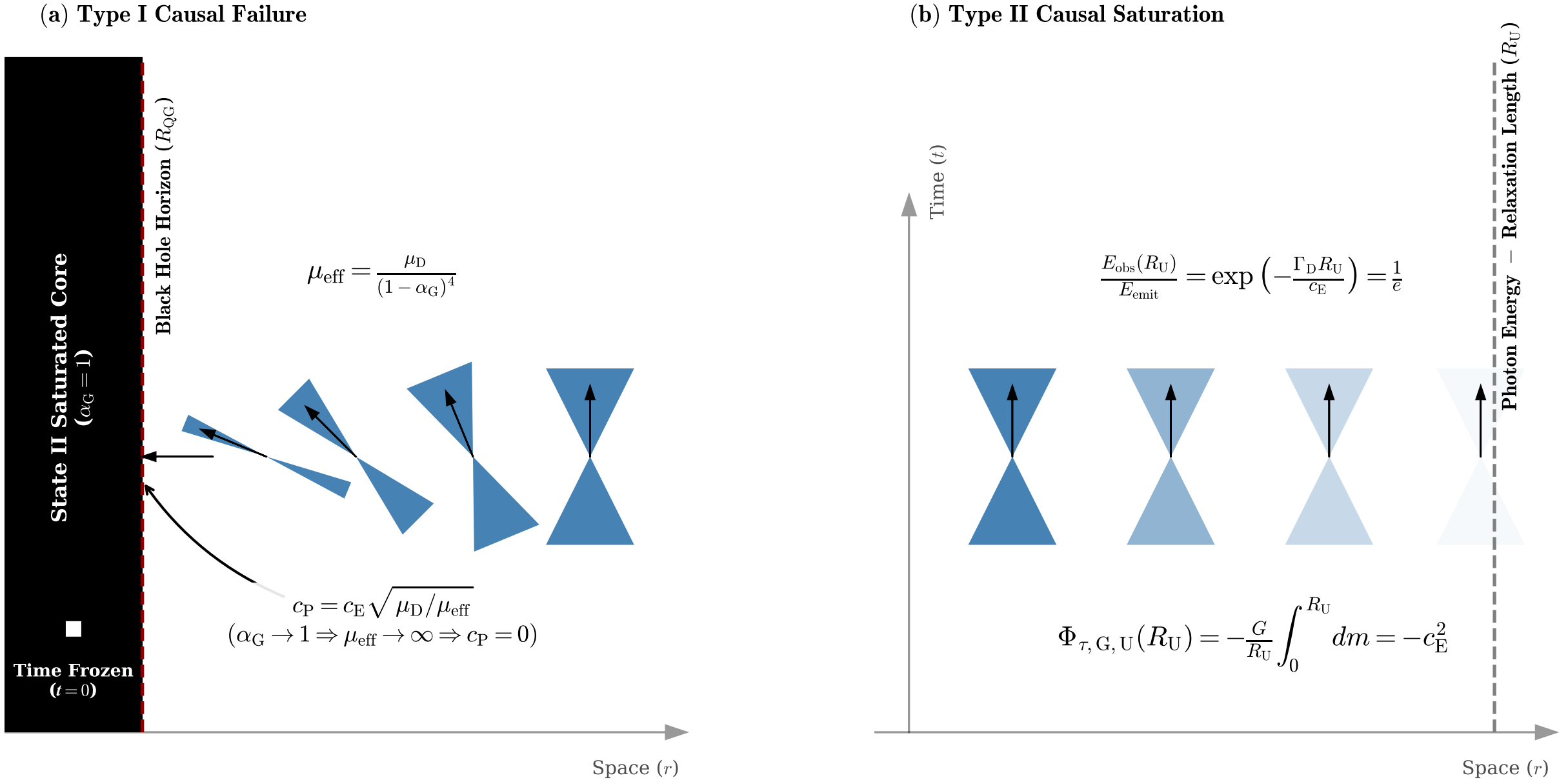

(a) Type I Causal Failure
State II Saturated Core
($\alpha_G = 1$)
Time Frozen
($t = 0$)
Black Hole Horizon (R_{QG})
$\mu_{eff} = \frac{\mu_D}{(1-\alpha_G)^4}$
$c_P = c_E \sqrt{\mu_D / \mu_{eff}}$
($\alpha_G \to 1 \Rightarrow \mu_{eff} \to \infty \Rightarrow c_P = 0$)
Space (r)
(b) Type II Causal Saturation
Time (t)
$\frac{E_{obs}(R_U)}{E_{emit}} = \exp\left(-\frac{\Gamma_D R_U}{c_E}\right) = \frac{1}{e}$
$\Phi_{\tau, G, U}(R_U) = -\frac{G}{R_U} \int_0^{R_U} dm = -c_E^2$
Photon Energy = Relaxation Length (R_U)
Space (r)